

GreenStep Employee Sustainability Challenge

Project Report

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Contents

1	Project Overview	2
1.1	Problem	2
1.2	Purpose	2
1.3	Users	2
1.4	Constraints	2
1.5	Proposed Solution	3
1.6	Technology	3
1.7	Expected Benefits	3
2	Current Functionality & Features	3
2.1	Authentication and Onboarding	3
2.2	Action Logging	4
2.3	Dashboard, Streaks, and Badges	5
2.4	Teams and Admin Approval	6
2.5	Leaderboards	6
2.6	Challenges and Join-by-Code	7
2.7	Administrator Tools	7
3	To-Do List & Future Enhancements	8
3.1	Technical	8
3.2	Functional	9
3.3	Usability	9
3.4	Future Expansion	9

1 Project Overview

GreenStep Employee is a gamified, team-based web application that replaces manual, spreadsheet-based sustainability tracking with a fast digital platform. It was built as a CISC480 Senior Capstone project for the community partner Minnesota GreenStep, and extends prior human-centred design work produced by a CISC489 team [7].

1.1 Problem

Workplace sustainability challenges (such as Earth Month) are commonly tracked with shared spreadsheets. Employees record their actions in a grid, and an organizer manually tallies the results at the end of the period. This approach suffers from “spreadsheet fatigue”: data entry is tedious, there is no real-time feedback, engagement drops over the course of the challenge, and administrators find it difficult to aggregate and report on participation.

1.2 Purpose

The purpose of GreenStep is to replace the spreadsheet with a low-friction, gamified digital tracker. By giving employees immediate feedback—points, badges, streaks, and live leaderboards—the platform aims to sustain engagement throughout a challenge while making aggregation and reporting effortless for administrators.

1.3 Users

The system targets two primary user groups, consistent with the personas defined in the prior CISC489 design work [7]:

- **Employees** (e.g. the “Maria” and “Sarah” personas) — everyday participants who log sustainable actions, build streaks, and compete on team and individual leaderboards.
- **Sustainability administrators** (e.g. the “David” persona) — organizers who create challenges, review and approve team requests, and export participation data for reporting.

Division or department leaders are a secondary audience, served indirectly through team-based leaderboards that group participants by team.

1.4 Constraints

The project operated under several constraints that shaped its design:

- **Zero budget** — all infrastructure runs on free service tiers (Vercel, Neon, Clerk).
- **One semester** — a fixed, short development timeline.
- **Four-person team** — limited engineering capacity.
- **Self-reported data** — actions are not independently verified, by partner agreement.

- **No dedicated DevOps** — the solution must be low-maintenance and require no ongoing operational staffing.

1.5 Proposed Solution

GreenStep is a single team-based web application in which employees log actions to earn points, badges, and streaks; compete on team and individual leaderboards; and participate in one or more challenges. Administrators create challenges, approve team-creation requests, and export participation data as CSV. The entire system is delivered as one deployable unit, requiring no installation by end users.

1.6 Technology

The application is built on a modern, type-safe web stack:

- **Next.js 15** (App Router) with **TypeScript** for both the user interface and the API routes [1].
- **Tailwind CSS** for styling [4].
- **Clerk** for authentication and session management [2].
- **Prisma** as the ORM [3], backed by **Neon** serverless PostgreSQL [6].
- **Recharts** for the performance chart [5].
- **Vercel** for hosting and continuous deployment.

1.7 Expected Benefits

Relative to the spreadsheet workflow it replaces, GreenStep is expected to deliver faster action logging, gamified engagement that sustains participation, live leaderboards that create healthy competition, one-click administrative CSV export, and very low maintenance overhead thanks to its fully managed, serverless infrastructure.

2 Current Functionality & Features

This section describes the features that are implemented and working in the current build. Each major screen is shown as a captioned screenshot.

2.1 Authentication and Onboarding

Authentication is handled by Clerk. New users sign up, sign in, and complete a first-run onboarding flow that records their department and enrolls them in the default “global” challenge. The landing page (Figure 1) introduces the product and routes signed-in users straight to their dashboard.

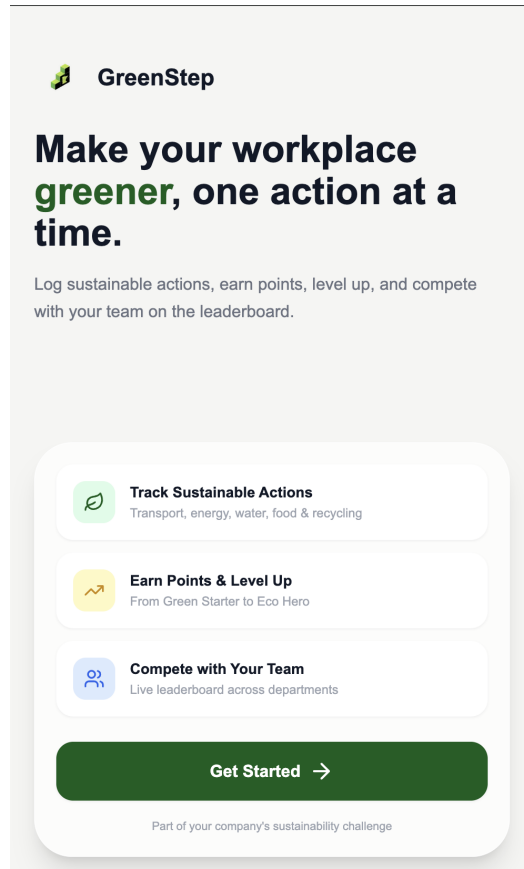


Figure 1: The GreenStep landing page, introducing the product and providing the sign-in entry point.

2.2 Action Logging

Authenticated users log sustainable actions across five categories —Transport, Water, Energy, Recycling, and Food. Each action carries a point value, and users may optionally attach a note or a photo for a small bonus. Users browse and search a catalog of pre-defined actions (Figure 2) and then confirm a specific action on its own page.

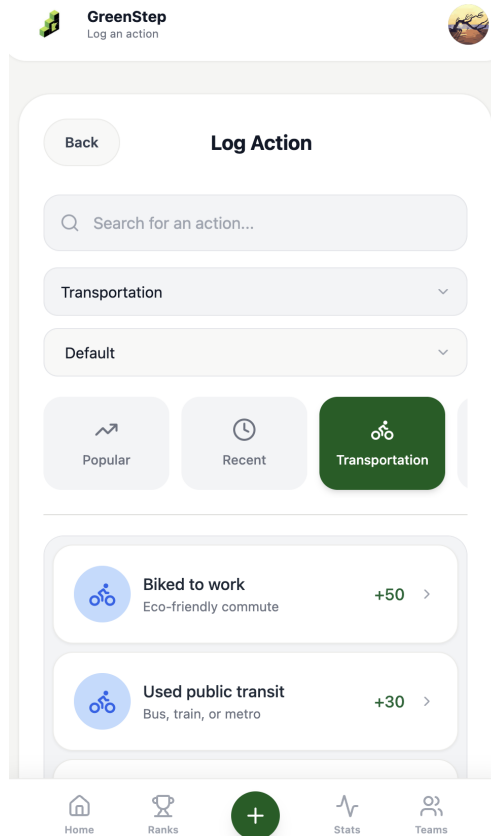


Figure 2: The action-logging screen, where users browse the five categories and select a specific action to log.

2.3 Dashboard, Streaks, and Badges

The dashboard (Figure 3) is the home screen. It shows the user's level and progress, current streak, recent personal activity, a community activity feed, and a performance chart comparing the user against the group average (rendered with Recharts). Daily engagement builds a **streak**, and users earn **badges** for milestones such as their first action, point thresholds, category-specific counts, and streak lengths.

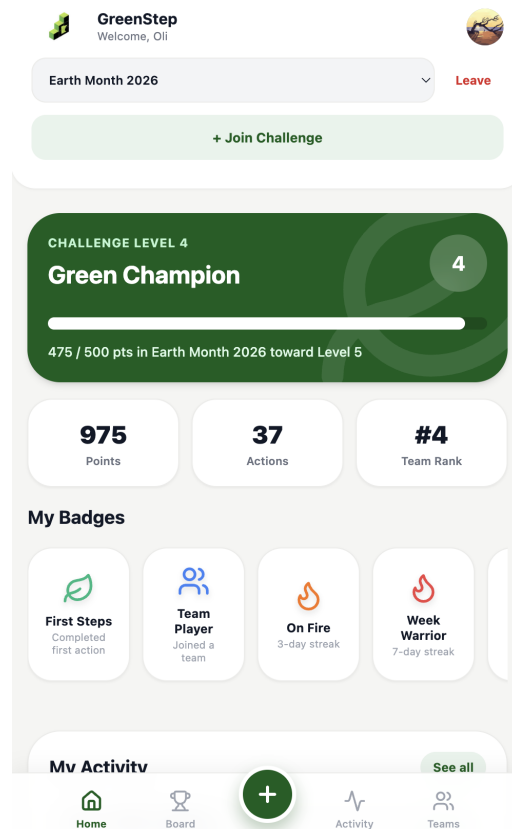


Figure 3: The main dashboard showing level progress, streak, recent activity, and the performance chart.

2.4 Teams and Admin Approval

Within a challenge, users join existing teams or submit a request to create a new team. Team-creation requests enter a **PENDING** state and must be approved (or rejected) by an administrator, giving organizers governance over team structure.

2.5 Leaderboards

GreenStep provides both **team** and **individual** leaderboards (Figure 4). Individual rankings offer several views—all-time points, rising performers, longest streaks, and category diversity—and both leaderboards can be filtered by category and time range.

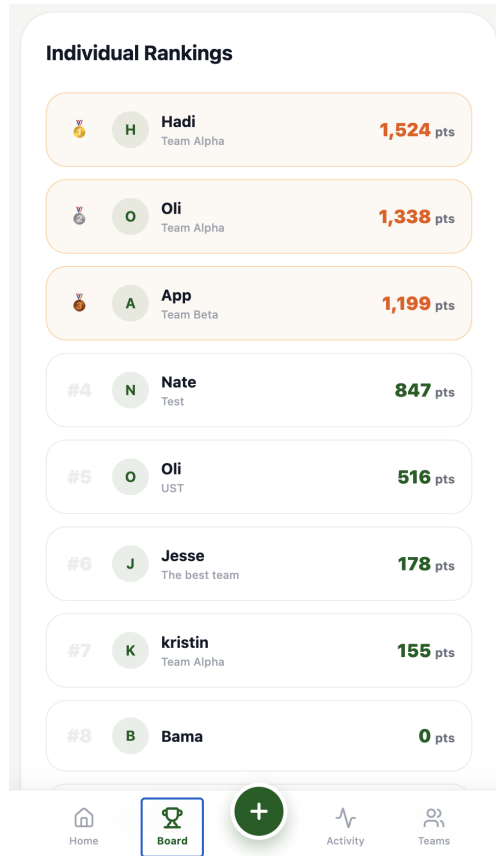


Figure 4: The leaderboard screen, showing team and individual rankings with category and time-range filters.

2.6 Challenges and Join-by-Code

Every user is auto-enrolled in a permanent global challenge and may join additional challenges using a unique join code. The active challenge scopes the dashboard, leaderboards, and activity feeds, and users can switch between challenges they belong to.

2.7 Administrator Tools

Administrators—identified by a Clerk `publicMetadata` role—have access to an admin screen (Figure 5) for creating challenges, reviewing and approving team requests, and exporting participation data to a CSV file for external reporting.

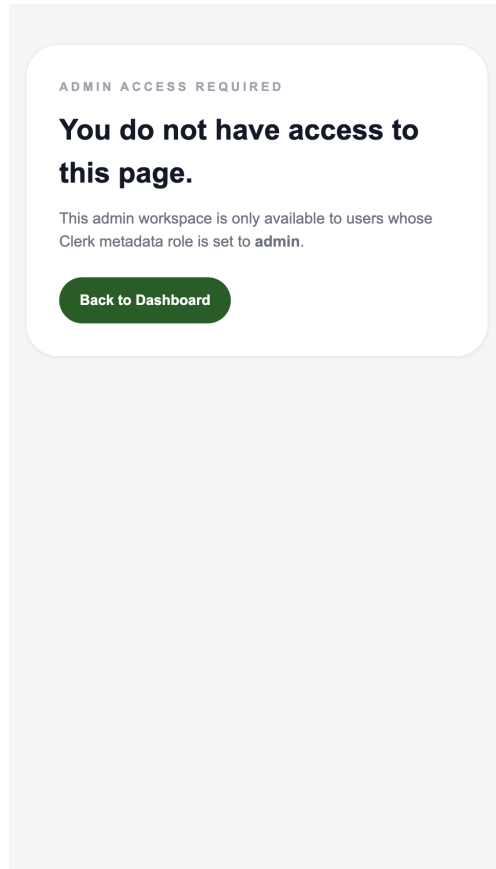


Figure 5: The administrator screen for creating challenges, reviewing team requests, and exporting CSV data.

3 To-Do List & Future Enhancements

The following items were out of scope for this semester or surfaced during development. They are grouped by aspect.

3.1 Technical

- **Automated testing.** The project currently relies on manual testing. Adding unit and end-to-end tests (e.g. for authentication, action logging, and leaderboard aggregation) and wiring them into CI would catch regressions automatically.
- **Further dashboard decomposition.** The main `DashboardClient.tsx` component was recently reduced (from roughly 2,300 to about 2,080 lines) by extracting its types, helpers, and the `useCountUp` hook into separate modules. The next step is to split it into per-tab components (Home, Leaderboard, Activity, Teams).
- **Database migrations.** The schema is currently applied with `prisma db push`. Moving to versioned `prisma migrate` migrations would provide a reproducible schema history for production.

3.2 Functional

- **Undo / edit a logged action.** Users cannot currently correct or remove an action after logging it.
- **Richer admin dashboard.** Expand the admin tools with participation analytics and per-team summaries.
- **Real challenge scheduling.** Honor the `startDate` and `endDate` fields to open and close challenges automatically.
- **User-selectable action dates.** Allow back-dating an action; actions are currently timestamped automatically.

3.3 Usability

- **Reminders and notifications.** Nudge users to maintain their streaks and re-engage lapsed participants.
- **Accessibility audit.** Conduct a formal review against WCAG guidelines (keyboard navigation, color contrast, screen-reader labels).
- **Environmental-impact feedback.** Show simple, category-based impact facts when an action is logged.

3.4 Future Expansion

- **Email digests.** Periodic summaries of team standings and personal progress.
- **Deeper analytics.** Trend analysis and reporting for organizers across challenges.
- **Multi-organization scaling.** Offer GreenStep as a multi-tenant service so other organizations can run their own challenges.

References

- [1] Vercel, “Next.js Documentation,” 2024. [Online]. Available: <https://nextjs.org/docs>
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- [6] Neon, “Neon Serverless Postgres Documentation,” 2024. [Online]. Available: <https://neon.tech/docs>

- [7] E. Adekoya, K. Alkhaleefa, and J. Dunlap, “Employee Sustainability Challenge App: Final Project Documentation,” CISC489 Topics in Human-Computer Interaction, University of St. Thomas, Dec. 2025.